



Baselines, Monitoring & AI-Summary Validation

Week 5 • Day 5 • Technical Infrastructure & Modeling

TPM Academy — TMD ships today



Day 5 Objectives

- Set **performance baselines** for Service Level Objectives (SLOs) + key operational signals
- Design a **three-layer monitoring plan** (on-call / operator / executive)
- Apply the **4-question test** to every alert
- Build an **AI-summary validation log** across all weeks
- Assemble + sign off the **full Technical Modeling Document (TMD)**



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin TMD Sections 1–4 on the wall
09:15 – 10:30	Block 1 + Activity 1	Performance baselines
10:45 – 12:00	Block 2 + Activity 2	Monitoring plan (3 layers)
13:00 – 14:15	Block 3 + Activity 3	AI-summary validation log
14:30 – 16:00	Block 4 + Activity 4 + Wrap	TMD integration + cross-review + sign-off

Block 1 — Performance Baselines

Where do we start measuring from?



Two Kinds of Baseline

Kind	When	Source
Existing-system	Brownfield (improving existing)	Production data
Expected-launch	Greenfield (new feature)	Estimated; named assumption

Greenfield discipline: state the assumption rather than skip the baseline.
"We expect ~200 reconcile-submits/day in the first 30 days, based on
dispatcher count × shifts/day × adoption."



Worked Baseline Table

Metric	Baseline	Source / assumption	Verify at
Reconcile p95 latency	~700ms	Sequence-diagram totals + 30%	Launch + 7d
Reconcile availability	99.5% (= TCD Section 4)	Inherits monolith	Launch + 30d
Reconcile-submits/day	~200	Dispatchers × shifts × adoption	Launch + 7d
Submission error rate	< 2%	Comparable features in app	Launch + 14d

"Verify at" dates make baselines real. Without them, baselines stay aspirational.

Activity 1 — Baselines

Quads • 35 minutes

- 5 min: list metrics needing baselines
- 5 min: classify (existing vs expected)
- 15 min: capture baseline values + assumptions
- 10 min: set "verify at" dates



5 + 5 + 15 + 10

Block 2 — Monitoring Plan

Three layers; fire before users notice



Three Layers of Monitoring

On-call alerts

Real-time. "Is something on fire?"
Pages or Slack.

Operator dashboard

Daily/weekly. "Is the system trending right?" TPM + eng lead.

Executive dashboard

Weekly/monthly. "Are we meeting SLOs?"
Leadership.

Three layers, three audiences, three rhythms.

The 4-Question Alert Test

1. **What does it mean?** Plain English
2. **What does on-call do?** Specific first step
3. **What's the threshold + window?** e.g., "5xx > 1% over 5 min"
4. **What's the severity?** Page / Slack / log

If you can't answer all four, the alert is noise — it'll be muted within a sprint.



Dashboard Right-Sizing

- **3–5 alerts** per feature. More = noise.
- **< 6 charts** on the operator dashboard. More = nobody reads.
- **< 4 charts** on the executive dashboard. More = leadership tunes out.

Track "mute reasons" on the operator level — captures the alert-tuning conversation.

Activity 2 — Monitoring Plan

Quads • 40 minutes

- 15 min: 3–5 on-call alerts (4-question test each)
- 15 min: operator dashboard (< 6 charts)
- 10 min: executive dashboard (< 4 charts)



15 + 15 + 10

Block 3 — AI-Summary Validation Log

The new discipline; cumulative across Weeks 2–5



Why AI-Summary Validation Matters

Failure	What happens
Wrong number	"p95 = 800ms" when chart shows 1200ms
Wrong direction	"improving" when trend is flat
Wrong attribution	Explains a spike with the wrong cause
Missing context	Headline summarized; relevant footnote missed

The TPM job: validate AI summaries against source data. Document the result.



Validation Status Values

- **Cross-checked all citations** — every claim verified (best)
- **Spot-checked** — N claims sampled and verified
- **Adopted with rationale** — correctness self-evident or cheap to verify
- **Pending** — used as input but not yet validated; track to closure
- **Rejected** — output was wrong; cut

The log is **cumulative** across the academy weeks. The next TPM should be able to audit it.



Worked Validation Log Entries

#	Purpose	Source	Validation
1	Strategy brief themes (Wk 2 D4)	14 tickets + 3 interviews	Cross-checked citations; 1 hallucinated; cut
2	Architecture critique (Wk 4 D1)	TCD Section 1	Adopted 2/3 objections
3	API critique (Wk 5 D3)	TMD Section 3	Adopted 4/5 issues
4	Cloud risk scan (Wk 5 D2)	TMD Section 2	All 3 valid; 1 added to TCD risks
5	Sequence critique (Wk 5 D4)	TMD Section 4	Flagged missing audit publish; added

Activity 3 — AI-Summary Validation Log

Quads • 40 minutes

- 15 min: inventory all AI uses across Weeks 2–5
- 15 min: capture validation row per use
- 5 min: handle "Pending" — validate, defer, or accept risk
- 5 min: sanity check for unlogged uses



15 + 15 + 5 + 5

Block 4 — TMD Integration + Sign-Off

Read top to bottom; cross-review; ship



The Friday TMD Rubric

Dimension	Weight
Data model clarity	20%
Cloud topology realism	15%
API contract sharpness	20%
Sequence model completeness	20%
Baselines + monitoring	15%
AI-validation discipline	10%

Score thresholds: ≥ 3.0 ships as-is. 2.0–2.9 with named gaps. $< 2.0 \rightarrow$ facilitator review.

Activity 4 — Integration + Sign-Off

Quads • 45 minutes • Wrap

- 10 min: solo read-through
- 5 min: pool issues
- 20 min: cross-review with rubric
- 10 min: adopt / defer / reject + sign off



10 + 5 + 20 + 10 • 15-min wrap



Week 5 Wrap

What we shipped

- Data model (Section 1)
- Cloud topology + rough order of magnitude (ROM) cost (Section 2)
- API contract (Section 3)
- Three sequence diagrams (Section 4)
- Baselines + monitoring + AI log (Section 5)
- Integrated TMD

What opens Week 6

- **Stakeholder Alignment & Negotiation**
- Stakeholder mapping
- Communicating tech trade-offs
- Negotiating priorities & roadmaps
- Technical Concept Document (TCD) Section 6 becomes the negotiation list

Bring to Monday: Your Product Requirements Document (PRD) + TCD + TMD. They are inputs to stakeholder conversations.

Questions & Reflection

Where in your TMD did the "I don't actually know" answer surface most strongly?